

PWV measurements with a GNSS receiver for the StarDICE experiment

Kélian SOMMER

Postdoctoral Research Associate at Steward Observatory

Atmosphere characterization workshop - April 1-3, 2025 - La Serena, Chile



UNIVERSITÉ DE
MONTPELLIER



Contact: sommerk@arizona.edu

Basic principles

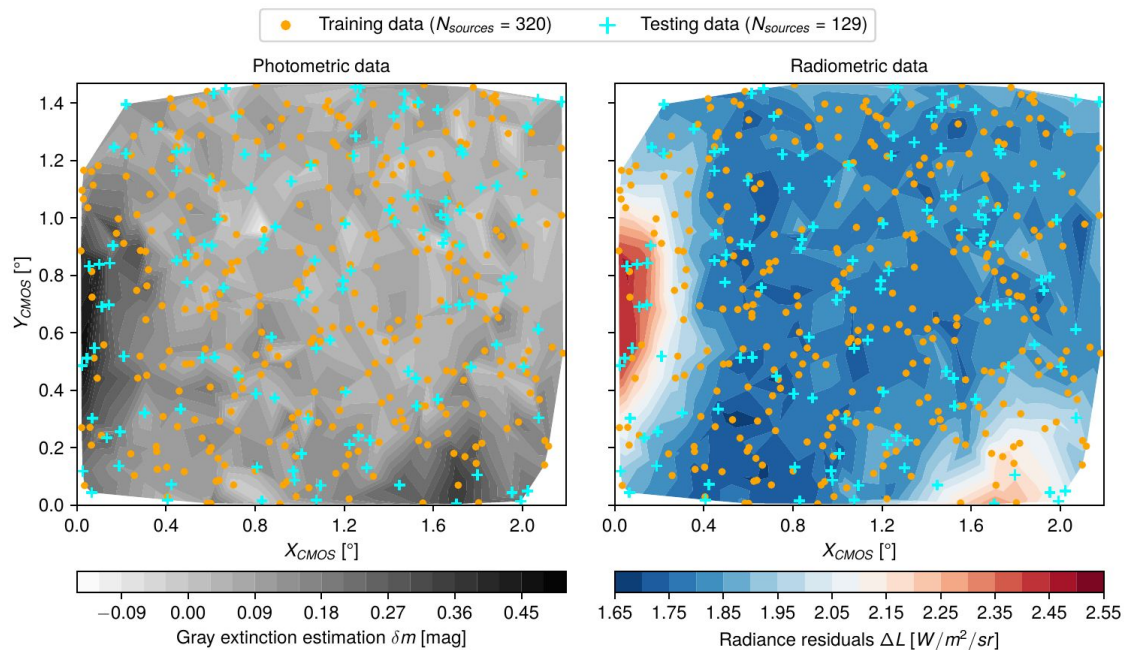
- **GNSS signal propagation delays:**
 - **Ionosphere:** Delayed by free electrons (corrected using 2 or more frequencies).
 - **Troposphere:** Delayed by air density.
- The **Zenith Total Delay (ZTD)** is derived from PPP and includes:
 - **ZHD (dry component):** calculated using models like Saastamoinen, based on latitude, atmospheric pressure, temperature and site height.
 - **ZWD (wet component):** can be derived to compute the PWV
- Requires a posteriori satellites clock and orbits

$$\text{ZTD} \approx 90\% \text{ ZHD} + 10\% \text{ ZWD}$$

$$\text{ZTD} = \text{ZHD} + \text{ZWD}$$

Motivation

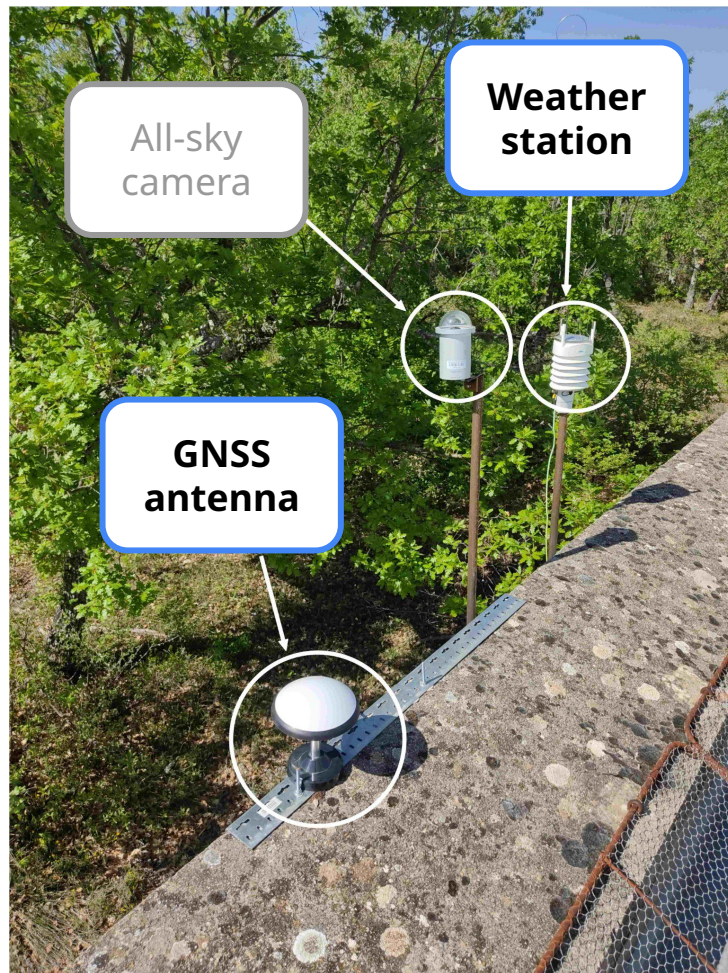
- OHP is a really wet site
⇒ 12 mm yearly average
- Provide PWV local measurements at zenith used for:
 - Near-UV to near-IR transmission modelling
 - Long-wave infrared thermal camera (centered around $10.8\text{ }\mu\text{m}$) → build a map of gray extinction in the telescope line of sight



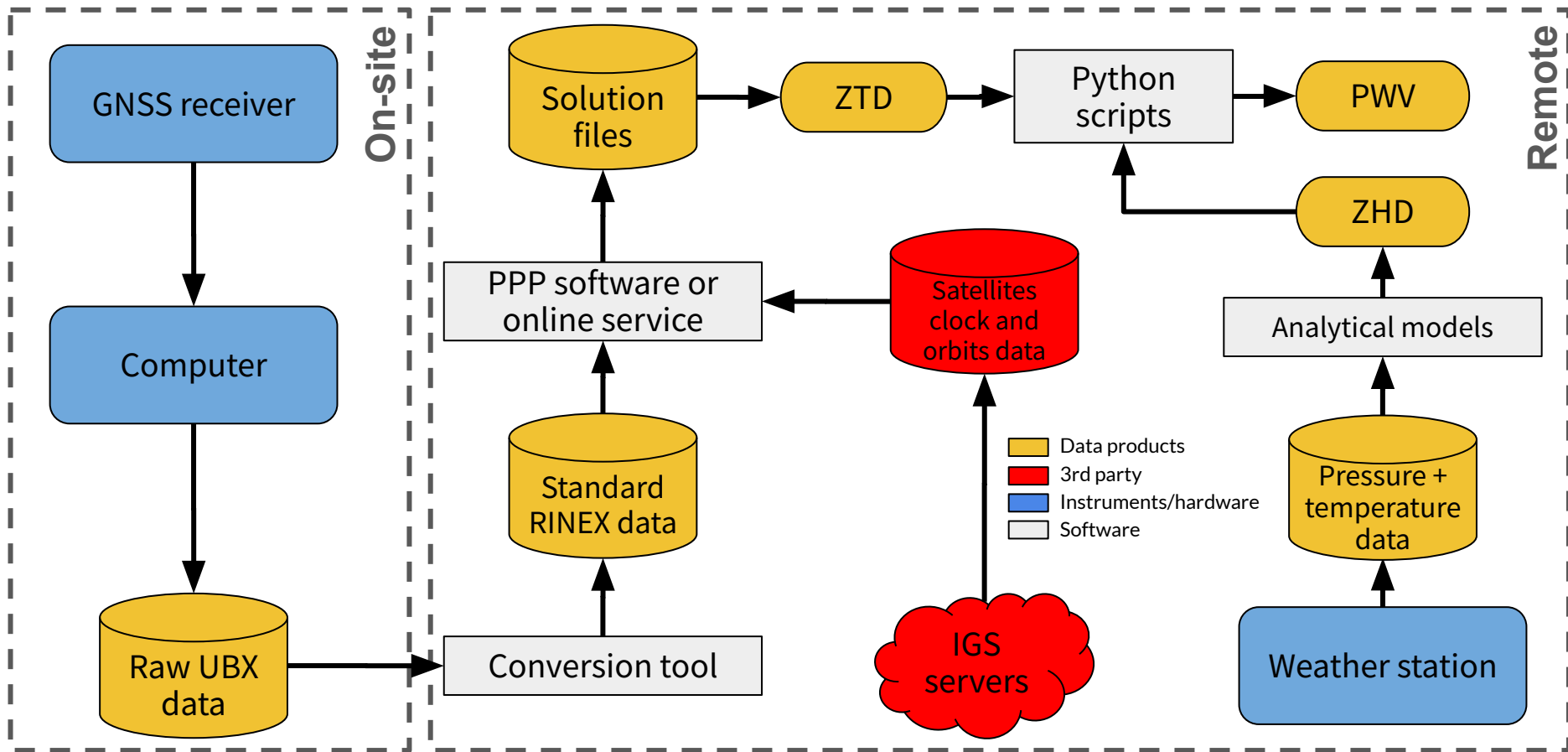
Required hardware

Item	Reference	Price \$
GNSS receiver	u-blox ZED-F9P	~200
Antenna	Tallysman VSP6037L	~800
Weather station	Vaisala WTX536 or any sensor with barometric pressure + temperature reading abilities	~200-4000
Computer w/ storage	HP Mini Computer or any	~80-300

1 day of raw data @ 0.1 Hz = 250MB

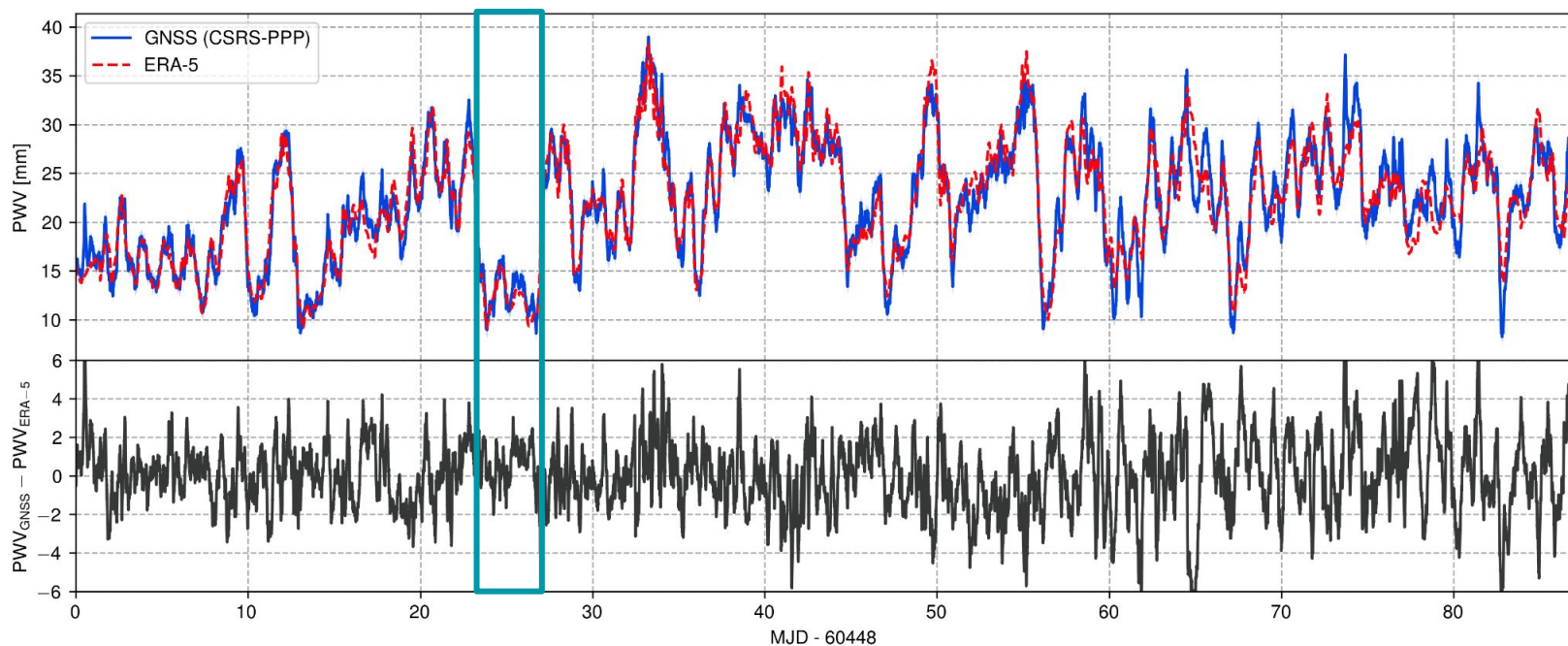


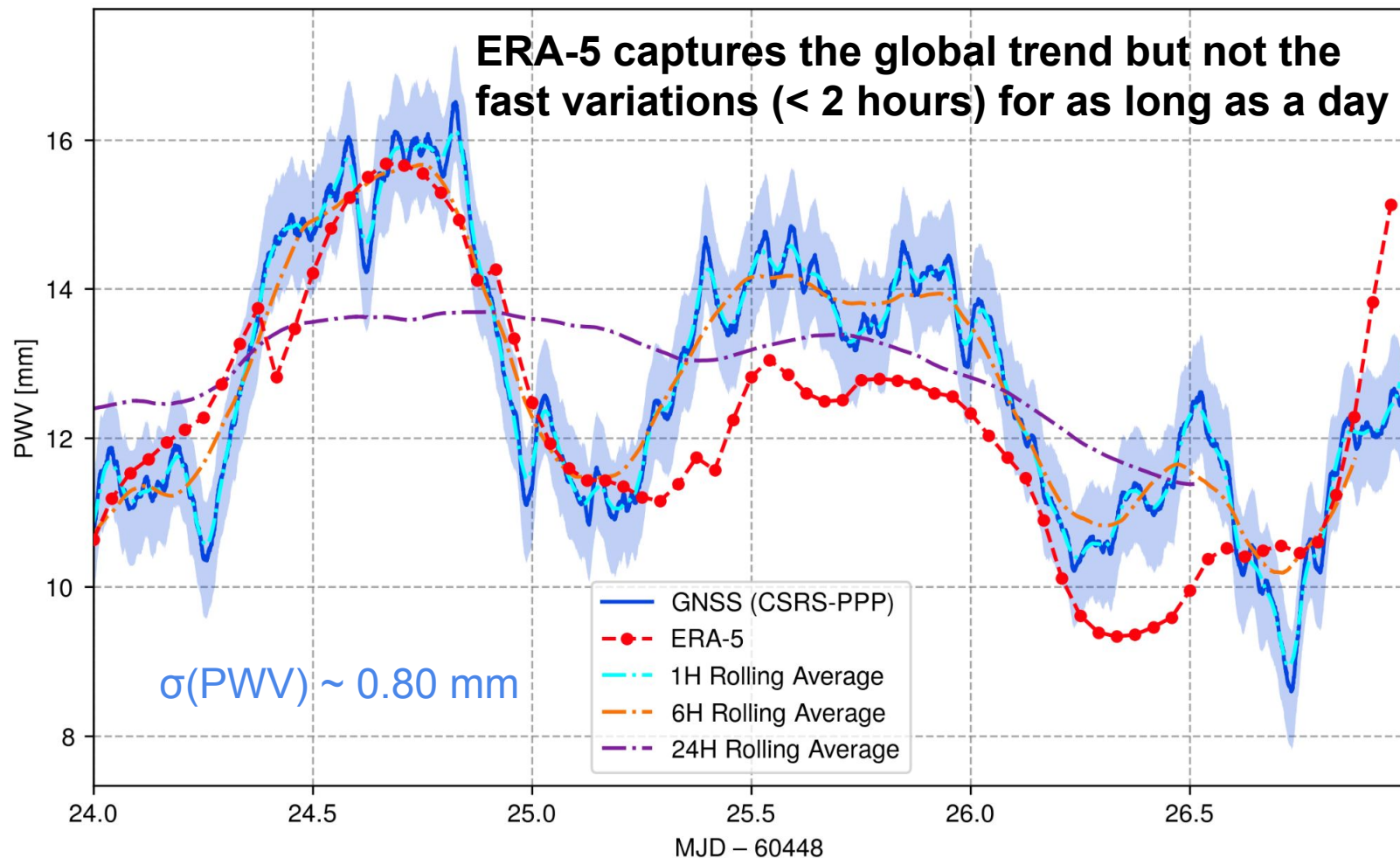
From raw data to final PWV measurements



Comparison of GNSS data vs ERA-5 data

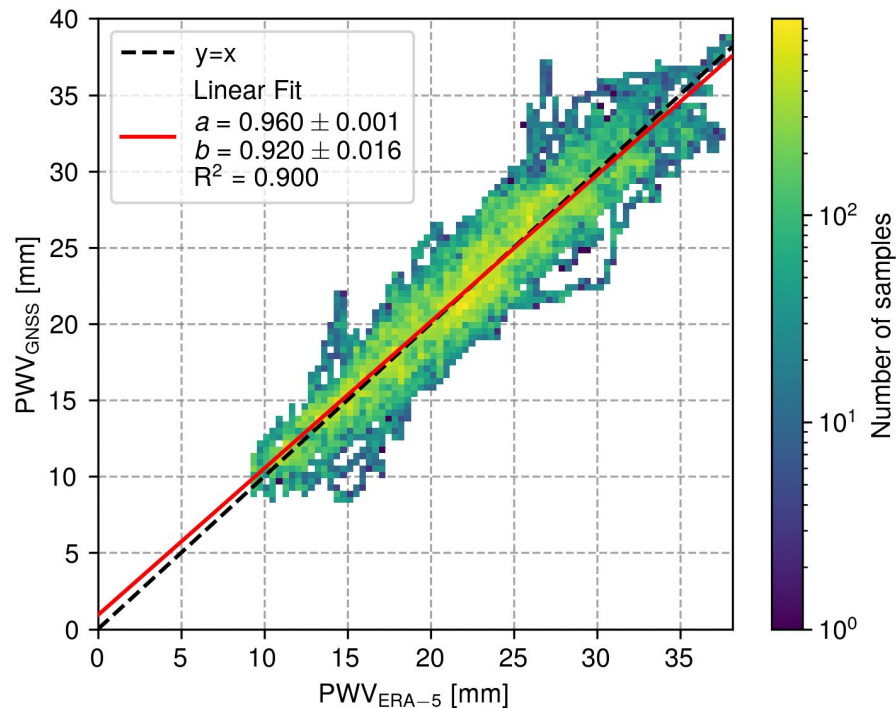
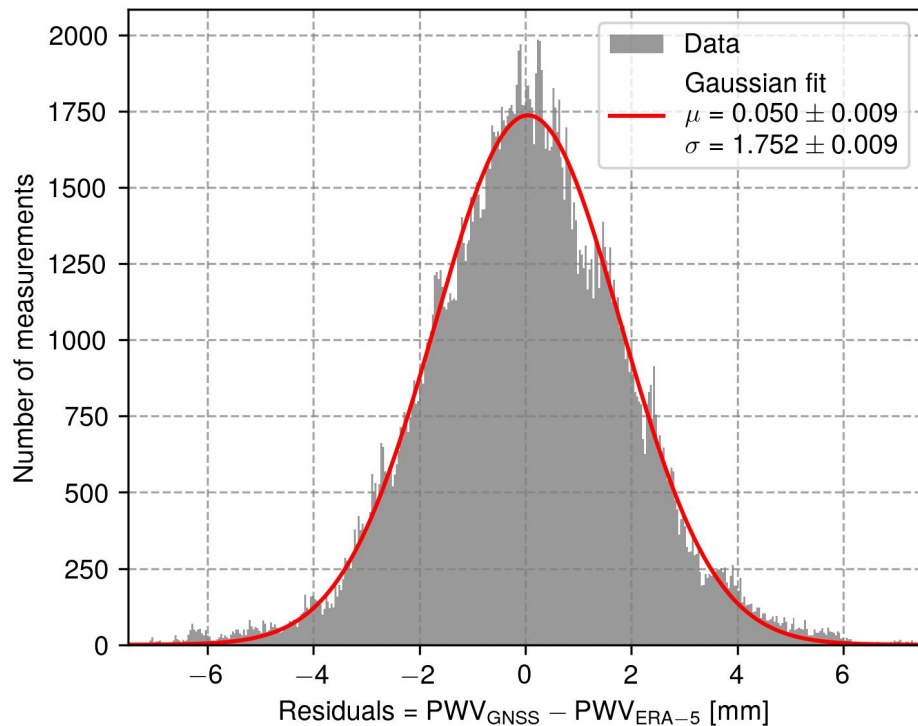
- 3 months collected at Observatoire de Haute-Provence (IAU code = 511)
- 15th May to 15th August 2024 = **very high humidity season at OHP**
- 30 seconds sampling for GNSS vs 1 hour (= max.) for ERA-5
- GNSS processed using free online service from Canada (CSRS-PPP)





Comparison of GNSS data vs ERA-5 data

- Practically no bias = 0.05 mm
- RMS of residuals $\sim 2\times$ the individual GNSS measurement uncertainties



Conclusions

- **Pros:**

- High-accuracy:
 - Temporal resolution: per exposure or lower (vs ERA-5 & MERRA-2 = 1 hour)
→capture spurious variations on short time-scales (<1 min)
 - Located on the observation site→no averaging over a large area like satellite data (e.g., ERA-5 ~ 31 km with $0.25 \times 0.25^\circ$; MERRA-2 ~ 50 km with $0.5 \times 0.625^\circ$)→precision issues with Vera Rubin Observatory location?
 - <1 mm of uncertainty per sample
- Cheap and easy to install/operate:
 - Available off-the-shelf components
 - <1500\$ (excluding weather station)
 - Run continuously without interruption

- **Cons:**

- Another instrument
- Requires a nearby or dedicated weather station providing barometric and temperature data
- If not CSRS processing, PPP softwares are hard to use
- No angle dependence capabilities

Discussion

- Appears to be a good trade-off solution (between cost, ease of operation and accuracy) to be sensitive to local and fast variations of PWV compared to satellite-based remote sensing observations
⇒ tried and tested at KPNO (Wood-Vasey et al. 2022; ApJ), Cerro Tocco for CMB experiments (Sugiyama et al. 2024; 2308.12632) and OHP for StarDICE.
- **Questions:**
 - Could GNSS PWV measurements be a “strong prior” to atmosphere transmission modelling relying on standard spectrophotometric stars spectra obtained with AuxTel ?
 - Would this kind of instrument be valuable/complementary to AuxTel?

Backup

PWV determination formula with GNSS information

Saastamoinen 1972 model:

$$ZHD = \frac{2.2767 \times P_s}{1 - 0.00266 \cos 2\varphi - 0.00028 \times h_0}$$

P_s is the surface pressure (in hPa)

φ is the latitude (in radians)

h_0 is the ellipsoidal height (in km)

$$ZWD = \boxed{ZTD} - ZHD$$

given by GNSS/PPP data

$$PWV = \frac{10^6}{\rho_w R_v \left[\frac{k_3}{T_m} + k'_2 \right]} \cdot ZWD$$

$\rho_w = 1000 \text{ kg/m}^3$, is the density of liquid water

$R_v = 461.495 \text{ J/kg/K}$ is the water vapor gas constant

k'_2 and k_3 are the atmospheric physical parameters having typical empirical values of $22.13 \pm 2.20 \text{ K/hPa}$ and $(3.739 \pm 0.012) \times 10^5 \text{ K}^2/\text{hPa}$

T_m is the atmospheric weighted average temperature (in K)

T_s is the surface temperature (in K)

$$T_m \simeq 70.2 + 0.72 T_s$$

Uncertainty on PWV

$$\sigma_V = \sqrt{\left(\frac{\sigma_{\text{ZTD}}}{Q}\right)^2 + \left(\frac{\sigma_{\text{ZHD}}}{Q}\right)^2 + \left(V \frac{\sigma_Q}{Q}\right)^2}$$

$$\sigma_{\text{ZHD}} = \sqrt{\left(\frac{2.2767 \sigma_{P_0}}{f(\lambda, H)}\right)^2 + \left(\frac{P_0 \sigma_c}{f(\lambda, H)}\right)^2}$$

$$\sigma_Q = 10^{-6} \rho_w R_w \sqrt{\left(\frac{\sigma_{k_3}}{T_m}\right)^2 + \sigma_{k'_2}^2 + \left(k_3 \frac{\sigma_{T_m}}{T_m^2}\right)^2}$$

$$\sigma_V = \sqrt{\left(\frac{\sigma_{\text{ZTD}}}{Q}\right)^2 + \left(\frac{2.2767 \sigma_{P_0}}{f(\lambda, H) Q}\right)^2 + \left(\frac{P_0 \sigma_c}{f(\lambda, H) Q}\right)^2 + \left(V \frac{\sigma_Q}{Q}\right)^2}$$

$$Q = 10^{-6} \rho_w R_w \left(k'_2 + \frac{k_3}{T_m}\right)$$

- $\sigma(P_0) \sim 0.12$ hPa
- $\sigma(T_m)/T_m \sim 2\%$
- k'_2 and k_3 are estimated from laboratory experiments with sub-percent accuracy
- The uncertainties of the density of liquid water (ρ_w) and the specific gas constant for water vapour (R_w) are 0.002 kg/m^3 and $0.008 \text{ J/kg/K} \rightarrow$ negligible

$$\Rightarrow 0.45 \leq \sigma(\text{PWV}) \leq 0.95 \text{ mm}$$

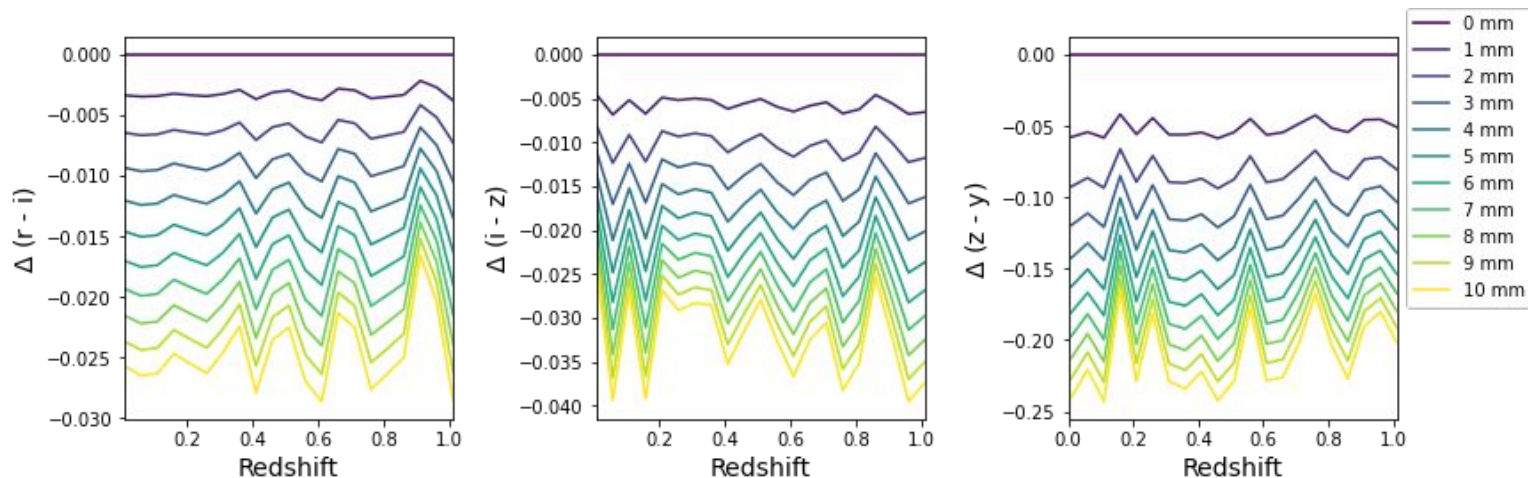
PPP services and softwares

https://gssc.esa.int/navipedia/index.php/PPP_Systems

Sensors 2018, 18, 580; doi:10.3390/s18020580

Parameter	APPS	CSRS-PPP	magicGNSS	gLAB	POINT	RTKLIB
Version	GIPSY 6.4	1.05	N/A	5.0.0	N/A	2.4.3
Mode of calculation	Static/kine-matic	Static/kine-matic	Static/kine-matic	Static/kine-matic	Static/kine-matic	Static/kine-matic
Constellation	GPS	GPS,GLO	GPS,GLO, Galileo,BDS	GPS, GLO,Galileo	GPS,GLO	GPS,GLO, GPS+GLO
Frequency	L1,L2	L1,L2	L1,L2	L1,L2	L1,L2	L1,L2
Type of observation	Code and phase	Code and phase	Code and phase	Code and phase	Code and phase	Code and phase
Antenna model	Not taken into account	Taken into account	Not taken into account	Taken into account	Taken into account	Taken into account
Frame of reference	ITRF2008	ITRF2008	ITRF2008	ITRF2008	ITRF2008	ITRF2008
Orbits and clocks of satellites	JPL final	IGS final	GMV Rapid, IGS Rapid, IGS final	IGS final	IGS final	IGS final
Cut-off angle	10°	10°	10°	10°	10°	10°
Mapping Function	GMF	GMF	GMF	NMF	NMF	NMF

Effect of PWV on LSST *rizy* colors for SNe Ia



$$\Delta \text{PWV} = 1 \text{ mm} \Rightarrow \Delta \text{color} \geq 2 \text{ mmag}$$